Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

768
-13 -1.66%
Nov 30th, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

886
-23.00 -2.53%
Nov 30th, 2022

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

656
6 0.92%
Nov 30th, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

101.00
0.35 0.35%
Nov 30th, 2022

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

108.80
-0.30 -0.27%
Nov 30th, 2022

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

842
-8 -0.94%
Week Ending Nov 25th, 2022

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2301 (Jan) RMB/t (3pm close)

769.00
-1.50 -0.19%
Nov 30th, 2022 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
December 22 USD/dmt

99.40
-0.55 -0.55%
Nov 30th, 2022 (5.30 pm Print)

SHFE Rebar
RB2301 (Jan) RMB/t

3748
-17 -0.45%
Nov 30th, 2022 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

19.14
-0.16 -0.80%
Nov 29th, 2022

C5, W. Australia - Qingdao USD/t

8.50
-0.64 -6.96%
Nov 29th, 2022

Steel Price

Steel Rebar (China Domestic) RMB/t

3780
-60 -1.56%
Week Ending Nov 25th, 2022

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

131.54
1.34 1.03%
Week Ending Nov 25th, 2022

Steel Inventory in China
million tonnes

8.88
-0.27 -2.97%
Week Ending Nov 25th, 2022

Steel Price

Steel HRC (China Domestic) RMB/t

3880
20 0.52%
Week Ending Nov 25th, 2022

IRON ORE PORT STOCK INDEX (IOPI)

Nov 30th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	768	-13	-1.7%	749	866	643	1083	99.18	-1.43	-1.4%	99.19	121.18	82.29	159.06
IOPI58	58% Fe Fines	656	6	0.9%	674	734	493	907	85.09	1.07	1.3%	89.88	102.86	70.95	134.15
IOPI65	65% Fe Fines	886	-23	-2.5%	882	1026	811	1265	115.00	-2.72	-2.3%	117.34	144.18	108.78	187.25

IRON ORE SEABORNE INDEX (IOSI)

Nov 30th, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	101.00	0.35	0.35%	100.35	123.57	83.90	159.45
IOSI65	65% Fe Fines	108.80	-0.30	-0.27%	110.88	142.48	94.45	185.82

MARKET COMMENTARY

DCE iron ore futures declined by 0.32% today, the main contract closed 769. The traders' willingness to ship is more.The steel mills are purchase on demand. The overall trading sentiment of the market is general. PBF at Shandong port dealt 759-761 yuan/mt,decrease 10 yuan/mt; PBF at Tangshan port dealt 773-778 yuan/mt. The recent macro news is driven by good real estate news, and the market sentiment is optimistic. However, the purchasing managers' index of China's manufacturing industry was released in November today, showing a continuous decline, indicating that the pressure on economic operation has increased due to the frequent outbreaks of the epidemic. And from the current terminal data, the downstream demand is relatively light. Considering the fundamentals, according to the SMM survey, the operating rate of the blast furnace on November 30 increased by 0.44% to 91.97% month on month. It is estimated that the daily average hot metal output of the steel plant increased slightly, and the demand for iron ore was supported. To sum up, it is expected that iron ore prices will fluctuate in the short term.

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Nov 25th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	842	-8	-0.9%	865	1057	770	1405	105.17	-2.03	-1.89%	110.49	142.79	95.00	199.00

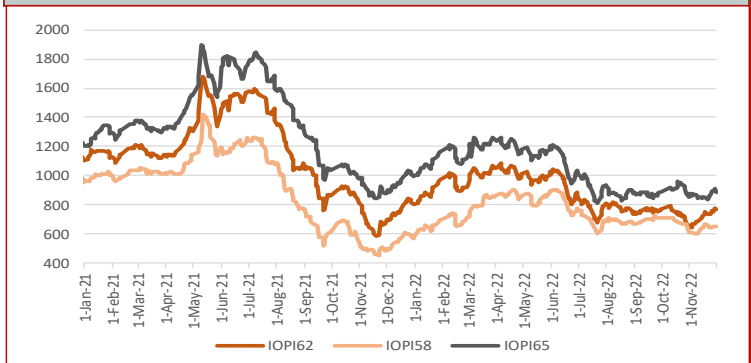
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Nov 25th, 2022				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	912	4.9%	779	1645	127.82	3.99%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	930	2.2%	780	1630	130.35	1.26%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	700	-0.7%	620	1310	98.11	-1.62%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	935	4.8%	800	1752	131.05	3.86%	117.19	272.32

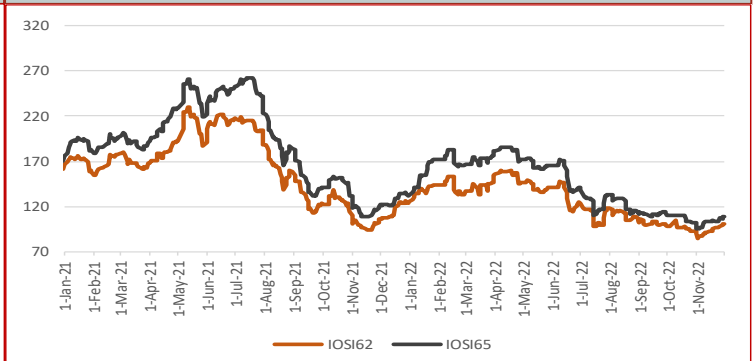
Week Ending Nov 25th, 2022	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	821.25	2.49%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =7.13488

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Nov 30th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD	August	September	October	November	MTD	QTD	YTD
IOPI62	62% Fe Fines	781	759	736	714	749	753	866	106.64	101.14	95.56	92.18	99.19	100.58	121.18
IOPI58	58% Fe Fines	685	697	683	635	674	676	734	94.05	93.56	89.34	82.39	89.88	90.81	102.86
IOPI65	65% Fe Fines	878	875	918	865	882	888	1026	120.31	117.19	120.08	112.37	117.34	119.25	144.18

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Nov 30th, 2022		CFR Qingdao, USD/dry tonne						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD
IOSI62	62% Fe Fines	110.64	101.12	95.80	93.05	100.35	102.24	123.57
IOSI65	65% Fe Fines	112.53	111.57	106.70	102.40	110.88	113.54	142.48

FREIGHT RATES

Nov 29th, 2022		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	8.50	-0.64	-6.96%	3.57	16.77
Tubarao - Qingdao	C3	19.14	-0.16	-0.80%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Nov 25th, 2022		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD	September	October	October	November	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	900	873	877	811	865	875	1057	118.50	112.75	109.81	100.83	110.49	112.84	142.79

IRON ORE INDEX PREMIUMS/DISCOUNTS

Nov 30th, 2022		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-112	-14.58%
IOPI65	65% Fe Fines	118	15.36%

Nov 30th, 2022		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	7.80	7.72%

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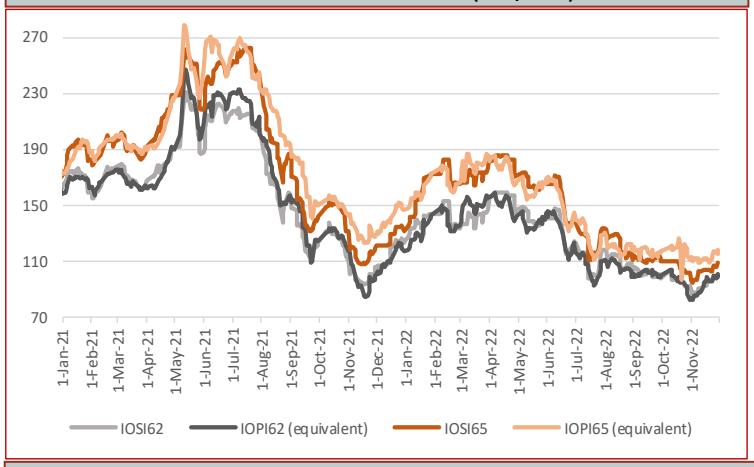
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Nov 30th, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	711	-13	-57
SIMEC Fines	638	-13	-130
PB Fines	738	-12	-30
Newman Fines	736	-13	-32
MAC Fines	718	-13	-50
Jimblebar Blended Fines	632	-13	-136
Carajas Fines	948	-13	180
Brazilian SSF	730	-13	-38
Brazilian Blend Fines	751	-13	-17
RTX Fines	651	-13	-117
West Pilbara Fines	680	-13	-88

Nov 30th, 2022	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	96.50	0.35	-4.50
SIMEC Fines	93.00	0.35	-8.00
PB Fines	97.25	0.35	-3.75
Newman Fines	100.15	0.35	-0.85
MAC Fines	97.25	0.35	-3.75
Jimblebar Blended Fines	89.65	0.35	-11.35
Carajas Fines	130.55	0.35	29.55
Brazilian SSF	104.75	0.35	3.75
Brazilian Blend Fines	106.40	0.35	5.40
RTX Fines	90.90	0.35	-10.10
West Pilbara Fines	95.25	0.35	-5.75

Nov 30th, 2022	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	656	6	0
FMG Blended Fines	664	6	8
Robe River	665	6	9
Western Fines	667	6	11
Atlas Fines	662	6	6
Yandi	650	6	-6

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	11.00	-1.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	19.00	-1.00
1% Alumina	High Fe Grade Al <2.25%	55.00	-5.00
	High Fe Grade Al 2.25-4%	11.00	-1.00
	Low Fe Grade Al <2.25%	28.00	5.00
	Low Fe Grade Al 2.25-4%	5.00	0.00
1% Silica	High Fe Grade Si <4%	20.00	-2.00
	High Fe Grade Si 4-6.5%	28.00	0.00
	Low Fe Grade	5.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	5.00	0.00
	High Fe Grade 0.115%<P<0.15%	5.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	0.50	0.00
	High Grade Fe 63 - 64%	2.50	0.25
	High Grade Fe 64 - 65%	2.50	0.25
	High Grade Fe 65 - 65.5%	2.50	0.25
1% Alumina	High Fe Grade Al <2.25%	2.50	0.25
	High Fe Grade Al 2.25-4%	0.75	-0.25
1% Silica	High Fe Grade Si <4%	2.50	0.25
	High Fe Grade Si 4 - 6.5%	0.25	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	3.00	0.00
	High Fe Grade 0.115%<P<0.15%	3.00	-0.25

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-20.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-10.00	0.00

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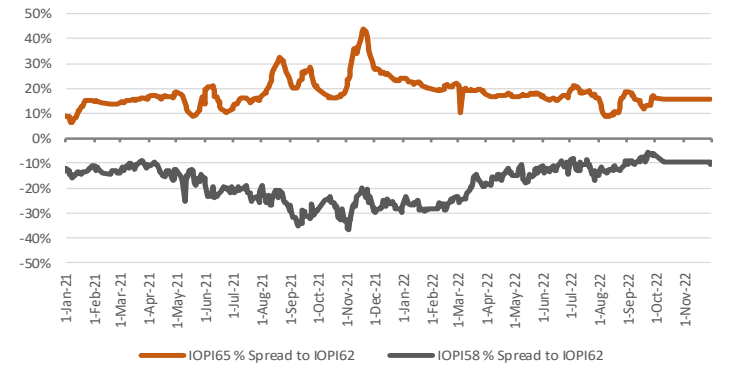
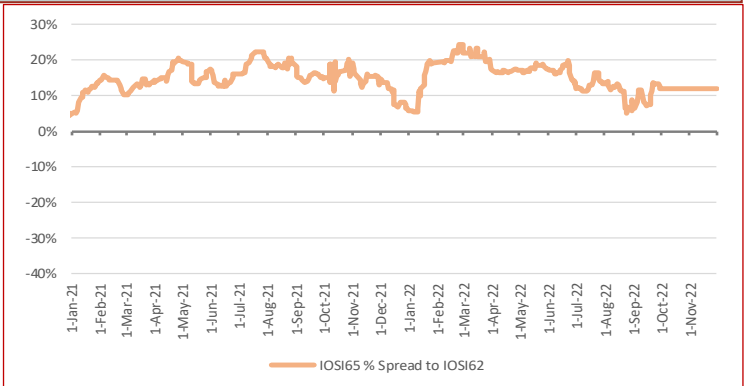
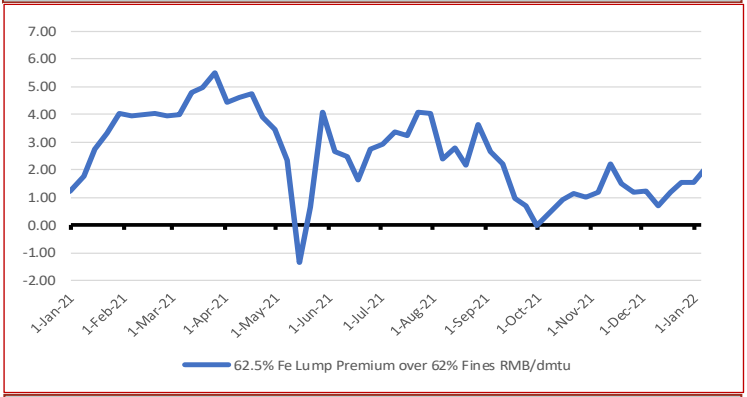
IRON ORE INDEX PREMIUMS/DISCOUNTS

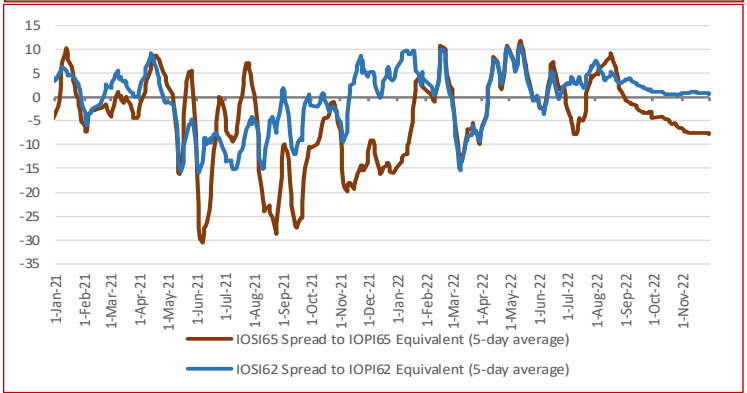




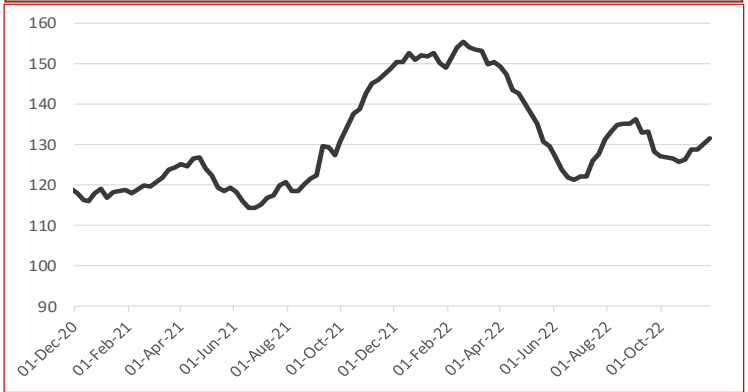
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



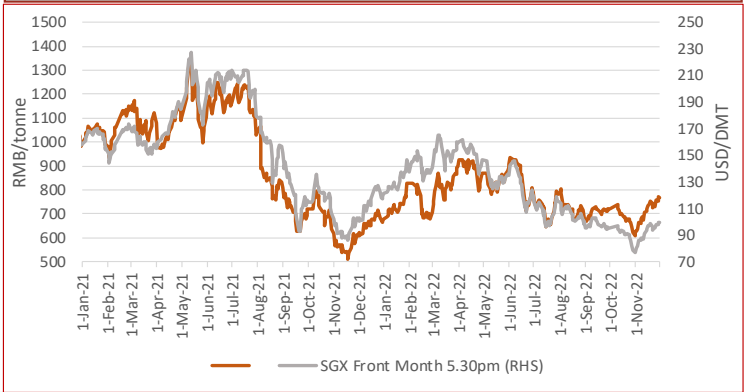
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

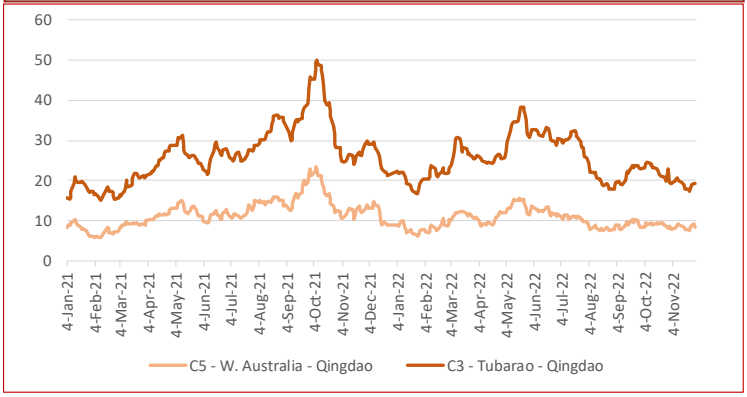
Week Ending Nov 25th, 2022

Province	This week	Change %	Low [2]	High [2]
Jingtang	13.34	1.83%	9.84	19.20
Qingdao	25.05	1.21%	9.41	26.24
Caofeidian	11.25	2.27%	9.05	16.29
Tianjin	9.18	-3.67%	7.14	12.97
Rizhao	18.37	2.45%	9.44	19.26
Total (35 Ports)	131.54	1.03%	98.80	155.39

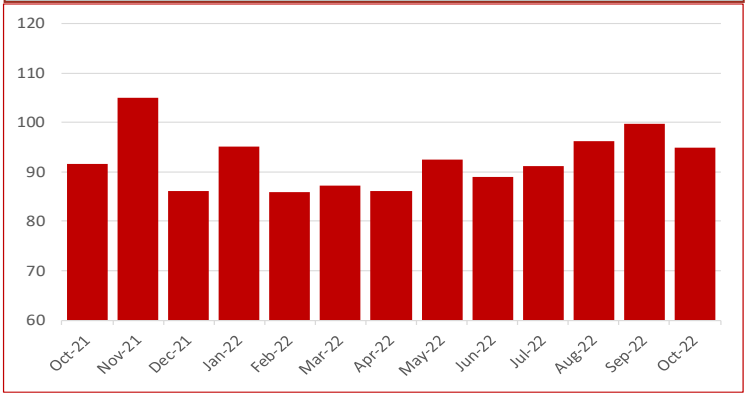
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Nov 30th, 3pm close			Nov 30th, 5:30pm		
Contract	I2301	Change	Change %	Dec. 22	Change	Change %
Closing Price	769.00	-1.50	-0.19%	99.40	-0.55	-0.55%
Vol traded ('000 lots)	56.49	-18.63	-24.80%	3.37	-1.74	-34.02%
Open positions ('000 lots)	59.28	-2.31	-3.75%	27.22	-1.20	-4.21%
Day Low	758.5	1.50	0.20%	99.85	1.65	1.68%
Day High	773.0	-7.50	-0.96%	101.35	-0.05	-0.05%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2022/11/25	Change	Change %
ReBar HRB400 ϕ18mm	3,780	-60	-1.56%
Wirerod Q300 ϕ6.5mm	4,290	-50	-1.15%
HRC Q235/SS400 5.5mm*1500*C	3,880	20	0.52%
CRC SPCC/ST12 1.0mm*1250*2500	4,340	-30	-0.69%
Medium & Heavy Plate Q235B 20mm	3,890	-20	-0.51%
GI ST02Z 1.0mm*1000*C	4,630	0	0.00%
Colour Coated Plate	7,500	0	0.00%

CHINESE STEEL MILL PROFITABILITY

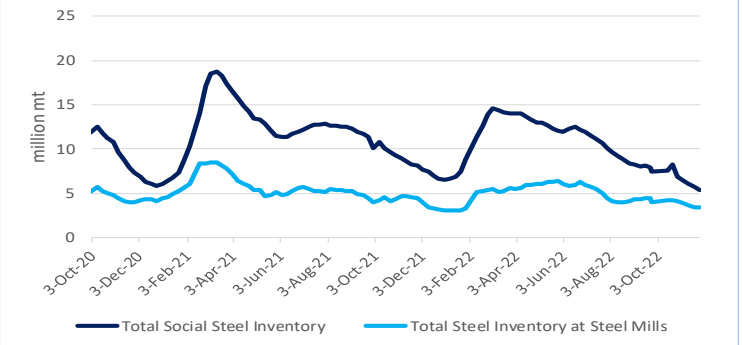
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	96.50	4	Mmi CFR Equivalent index for 1st Feb
Coke	2,850	110	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	2,910	0	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,506	47	Q234, incl. tax
Rebar cost - Blast furnace	3,996	54	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-206	-84	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	3,960	54	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-90	-54	based on Shanghai prices, incl. tax

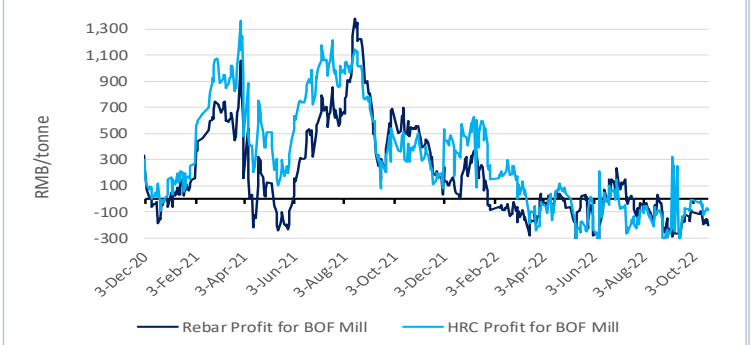
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

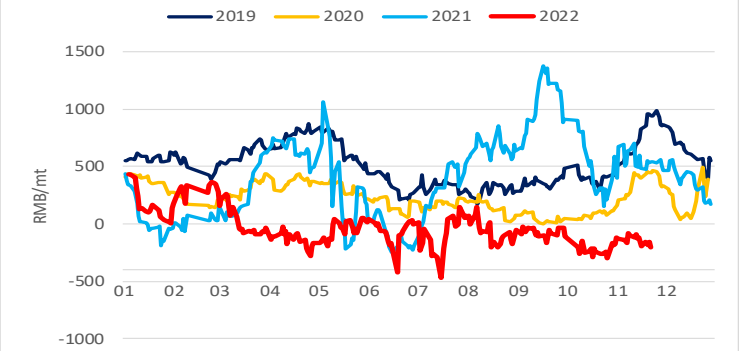
CHINESE STEEL INVENTORIES



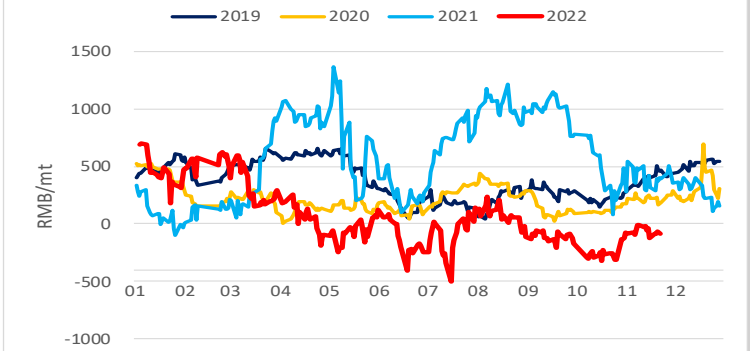
CHINESE STEEL MILL PROFITABILITY



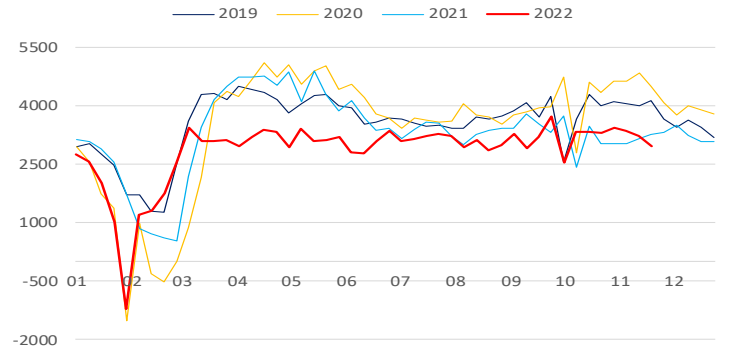
CHINESE STEEL MILL PROFITABILITY—Rebar



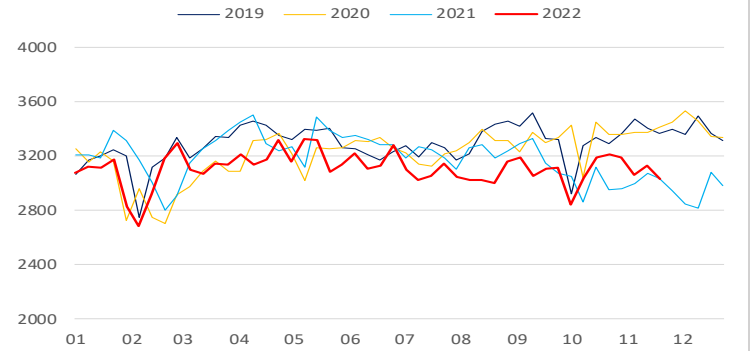
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



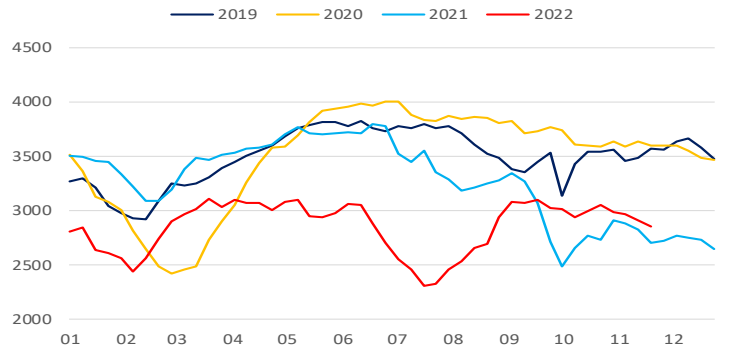
CHINESE STEEL CONSUMPTION—Rebar



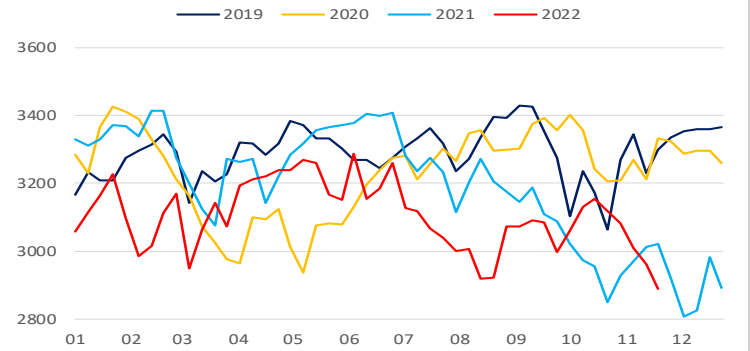
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Nov 30th, 2022	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Nov 30th, 2022	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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